A Domain Language for Specifying Controlled Methods

Staged Validation and Deterministic Compilation for Method Specification, Informed by BPL

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Contents

1 Abstract

This paper describes a small, tested **domain language for specifying controlled methods** — the **methods-paper exemplar** of the Research Project Template. Unlike a results paper, this manuscript's subject is the methodology itself: a controlled vocabulary, a unit system with dimensional safety, four staged validation gates, and a deterministic compiler, implemented in and described section by section in sec. 3. The domain language's vocabulary is informed by BPL (Biology Programming Language, [Bota Biosciences, 2026]), an upstream reference that encodes laboratory protocols as programs with biology-native types, staged validation, and deterministic compilation; this exemplar generalizes BPL's intent vocabulary and pipeline shape from wet-lab protocols to any controlled procedure.

A is a name, a set of typed parameters and resources, and an ordered, dependent set of steps — constructed directly as frozen Python dataclasses () rather than parsed from new text syntax. Every carries a unit that resolves to one of 18 controlled units across eight dimensions, and every step names one of 9 controlled-vocabulary intents (), executable on one of 3 backends. 4 staged gates — structural, semantic, plan, and target — validate a method before () deterministically schedules it with Kahn's algorithm [Kahn, 1962] and hashes the canonical plan with SHA-256.

We demonstrate the language on 2 worked example methods spanning both domains BPL's design targets and the domains it generalizes to: a manual wet-lab preparation (, 5 steps, target , plan hash) and an automated instrument-calibration procedure (, 4 steps, target , plan hash). Live re-compilation determinism check: **Yes**. Across both methods, 8 of 8 staged-gate evaluations pass. A demonstration provenance hash-chain () of length 3 verifies as **Yes**.

Contributions are **methodological** and **architectural**. On the methods side, we show that a controlled vocabulary expressed as typed dataclasses — not a parsed grammar — is sufficient to reproduce BPL's core safety properties (dimensional safety, staged validation, deterministic compilation) at a scope appropriate for a template exemplar. On the architecture side, the DSL is exercised by a zero-mock test suite under the repository's configured project coverage gate, generates 19 artifacts (1 figures, 7 data files, 11 reports) per pipeline run, and injects reproducibility metadata (configuration hash , build timestamp) into sec. 7.

Keywords: methods paper, domain-specific language, controlled methods, deterministic compilation, staged validation, dimensional analysis

2 Introduction

This serves as the **methods-paper exemplar** for the Research Project Template ecosystem: a manuscript whose subject is a methodology — here, a domain language for specifying controlled methods — rather than results produced by running one. The prose, the labelled figures, and the compiled-plan table are produced through the same auditable custody chain every exemplar in this template uses: tested functions in , a thin analysis script, and generated-variable-injected, multi-format rendering.

2.1 Why a methods paper needs its own DSL

A methods section in ordinary prose is ambiguous by construction: "add 10 mL of water, then mix" admits multiple readings of order, units, and what "mix" means operationally. BPL [Bota Biosciences, 2026] makes the case for laboratory protocols directly: free-text instructions admit multiple interpretations, unit errors and reagent mismatches surface only at the bench, and re-executing a protocol on a different operator or instrument introduces silent variation. Fowler frames the general remedy as a **domain-specific language** [Fowler, 2010]: a small, purpose-built notation whose vocabulary is restricted exactly to the concepts the domain needs, so that what can be written down is exactly what is intended.

2.2 What this exemplar borrows from BPL, and what it generalizes

BPL's architecture is a compiler pipeline — parse, semantic check, lower, schedule, execute, export — over a biology-native type system (units, dimensional analysis, MW-aware concentration), staged validation gates, and deterministic compilation with a stable plan hash. Three design choices carry over directly into :

1. **Intent over instruction.** BPL users write high-level intents (, ,); a compiler lowers them to backend-specific primitives. 's enum is the same idea, generalized: // name *what* happens, never *how* a particular backend performs it.
2. **Dimensional safety.** BPL's type system catches at compile time, not at the bench. implements the same guarantee with a small / system rather than a full unit library.
3. **Deterministic compilation.** Same source, same options, same plan hash. reproduces this with a canonical-JSON SHA-256 hash over a Kahn's-algorithm [Kahn, 1962] schedule.

What this exemplar does **not** carry over is BPL's text grammar and parser: a here is constructed directly as frozen Python dataclasses ( ), not parsed from source. This keeps the DSL's discipline in its typed, validated shape rather than in new concrete syntax — appropriate for a template exemplar's scope — while the controlled vocabulary, dimensional safety, and deterministic compilation generalize unchanged from wet-lab protocols to any controlled procedure, demonstrated in sec. 4 by one wet-lab-flavored method and one instrument-calibration method.

2.3 Template architecture context

The project sits on the repository's three pillars:

1. **library**: pure, side-effect-free dataclasses and functions — no plotting, no file I/O, and (with one declared logging exception) no imports. This purity is what makes the library forkable and trivially testable.
2. **framework**: a zero-mock suite that exercises every gate, the compiler, and the exporters against real fixtures covering both the success path and every gate-failure mode.
3. **knowledge base**: the correspondence with BPL's pipeline, the testing philosophy, and the operational rules that govern agents editing this tree.

2.4 The worked examples

We specify two methods with (): , an original — not copied from BPL's shipped examples — manual bench preparation in BPL's own domain, and , a non-biology controlled procedure mixing automated measurement with a human sign-off step. The second example exists specifically to demonstrate that the DSL's vocabulary generalizes beyond wet-lab protocols, as sec. 2 claims.

2.5 Reader's guide to the manuscript

- **sec. 3** ties each pipeline stage to its module in .
- **sec. 4** is artifact-centric: every reported number names the function or report file that produced it.
- **sec. 6** lists the controlled vocabulary and software environment.
- **sec. 7** records the artifact inventory and the exact commands to regenerate everything.
- **sec. 8** states scope and related work so the exemplar is not mistaken for a general-purpose protocol-execution system.

3 Methodology

The DSL is implemented as eight cooperating modules under , each corresponding to one stage of a BPL-inspired pipeline [Bota Biosciences, 2026]. This section walks the pipeline stage by stage, naming the function or class that implements each design decision so every claim below is directly checkable against .

3.1 Controlled vocabulary ()

A is one of 9 controlled intents — , , , , , , , , — and a is one of 3 execution backends — , , . encodes which kinds require an automated backend: only has no manual equivalent in this DSL's scope, so and accept every other kind. This is the domain-neutral generalization of BPL's protocol-level verbs (, ,): a step names *what* happens, never *how* a particular backend performs it.

3.2 Dimensional safety ()

Every resolves its to one of eight members (mass, volume, temperature, time, molar concentration, mass concentration, count, dimensionless) via , drawing from a controlled table of 18 unit strings. Molar concentration (,) and mass concentration () are separate dimensions rather than one shared "concentration" dimension, since converting between them needs a substance's molar mass that this DSL does not carry. raises the moment two quantities with different dimensions are combined — the concrete realization of BPL's design principle that "the type system catches at compile time, not at the bench." Temperature is tracked as its own dimension with no shared base unit (and are never auto-converted), since this DSL has no use for that conversion and an incorrect affine conversion is worse than refusing one.

3.3 Method model ()

A is a name, a version, a target, a tuple of declarations (anything a step reads from or writes to — generalizing BPL's /), a tuple of method-level s, and a tuple of s. Each carries a , a , a , its own parameters, an optional expected duration (must be a time), and a tuple of prerequisite s — the explicit DAG edges this section's compilation stage resolves. All four dataclasses are frozen; rejects malformed shapes immediately (empty names, self-dependency, a non-time) so structurally invalid methods cannot even be constructed, collapsing BPL's syntax gate into Python's own construction-time checks.

3.4 Staged validation ()

runs exactly 4 gates in fixed order, mirroring BPL's staged short-circuit (a syntax failure never reaches the plan gate):

1. — every is unique and every entry resolves to a real step.
2. — every attached to a resource, a method parameter, a step's expected duration, or a step parameter resolves to a known (catches the unit-vocabulary violation a frozen dataclass's cannot, since does not validate its unit eagerly).
3. — the step-dependency graph is acyclic, checked by attempting and catching .
4. — every step's target is compatible with the method's target (methods accept only steps; methods accept both; methods accept only steps) and every step's kind is executable on its assigned target.

If either of the first two gates fails, returns early with only those two results — and assume a structurally and semantically valid method and would otherwise report misleading secondary failures.

3.5 Deterministic compilation ()

first calls and raises (carrying every failed gate's issues) if any gate fails. On success, schedules the validated steps with Kahn's algorithm [Kahn, 1962]: repeatedly remove a step whose dependencies are all already scheduled, breaking ties by ascending so the same method always yields the same order — Python does not guarantee dict/set iteration order is stable for this purpose, so the tie-break is explicit, not incidental. The scheduled steps are then encoded as a canonical, sort-keys JSON payload and hashed with SHA-256 () into . Because the hash is computed purely from , , , and each step's ////scheduled — never from a wall-clock timestamp or a UUID — recompiling the same object always produces the same , which sec. 4 verifies live rather than asserts.

3.6 Export ()

A compiled renders to four formats, mirroring BPL's "CSV/XLSX worklists, workflow graphs" export surface at a scope appropriate for a template exemplar: (a numbered, human-readable worklist), / (machine-readable rows), (a showing scheduled order), and / (the exact canonical JSON the plan hash was computed over, so a reader can independently verify by re-hashing the exported file).

3.7 Provenance ()

orders three levels of trust — , , — generalizing BPL's audit model. extends an immutable hash-chain of s, each hashing its own // / [Merkle, 1987]; recomputes every record's hash and checks it against the recorded chain. This is a consistency check, not a cryptographic tamper-proof guarantee against an actor with write access to the whole stored chain: it detects in-chain tampering

(any record after a tampered one no longer matches) but cannot detect a chain rewritten from record zero, exactly the boundary BPL's own "hash-chained" (not cryptographically signed) audit model claims.

3.8 Zero-mock testing methodology

The project is governed by a strict zero-mock policy, evaluated by running during the build.

1. **Library tests** exercise every gate, the compiler, the exporters, and the trust module against real objects — ships one fixture per gate-failure mode (unknown dependency, duplicate step id, unknown unit, cyclic dependency, target mismatch) plus a linear-chain and a diamond-DAG method for scheduling tests. No , no , no .
2. **Script test** runs against a temporary output root and asserts that real worklist/CSV/Mermaid/JSON files, reports, and a real PNG figure are written.
3. **Determinism tests** compile the same object twice and assert equality live, rather than asserting against a hardcoded hash literal — a literal would silently stop testing the moment the compiler's hash input changed.
4. **Coverage gate**: CI enforces a >=90% statement-coverage gate on ; the live figure is tracked in .

4 Results

This section reports the compiled plans for both worked example methods. Every number below is produced by the methods analysis orchestrator (), which calls and from and writes , , and . Running the script regenerates every artifact this section references.

4.1 Compiled-plan summary

Table 1: Compiled-plan summary from , generated by for each of 2 worked example methods.

Method	Steps	Target	Plan hash (first 12 hex chars)
	5		
	4		

tbl. 1 shows (a manual, -target bench preparation) alongside (a mixed / instrument-calibration procedure) — the second example exists specifically to demonstrate the controlled vocabulary generalizing beyond wet-lab protocols, as sec. 2 claims.

4.2 Step-count comparison

fig. 1 plots the step count for each compiled method.

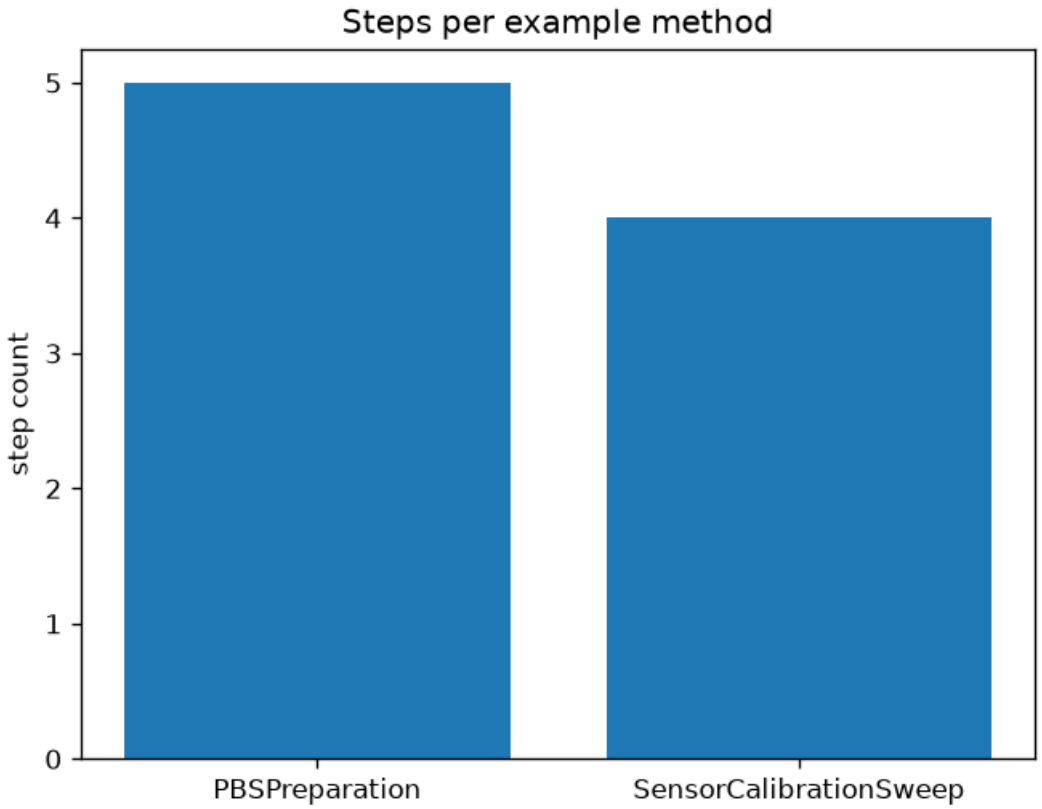

Figure 1: Steps per example method: a bar chart built from for each plan in , plotted by .

4.3 Validation gates

Across both methods, the analysis script tallies 8 of 8 staged-gate evaluations passing (× 2 methods × 4 gates each). Every gate result is written to — neither method in this manuscript is hand-picked to pass; both worked examples are constructed to satisfy the structural, semantic, plan, and target gates by design, since raises and halts the pipeline on any gate failure.

4.4 Determinism

Recompiling each example method twice and comparing values yields: **determinism check = Yes**. This is a live re-compilation comparison performed by at manuscript-build time, not a value asserted once and then transcribed — the same property sec. 3 claims for is checked again here, independently, against the live build.

4.5 Provenance demonstration

appends a 3-record demonstration hash-chain for one value () through → → tiers and writes the result of to : **chain verified = Yes**.

4.6 Validation

The results were validated through the zero-mock suite:

- **Library tests** assert exact gate outcomes, scheduling order, and plan-hash determinism against real fixtures, including one fixture per gate-failure mode.
- **Script test** runs against a temporary output root and confirms real worklist/CSV/Mermaid/JSON artifacts, reports, and a real PNG figure are written.
- **Manuscript-variable test** confirms every generated-variable name used in is emitted by .

All tests pass under the configured project coverage gate, with no mocks.

4.7 Discussion

The results confirm the pipeline end to end: both worked examples pass every staged gate, compile deterministically, and produce a stable plan hash across repeated builds. The same functions back the analysis script, the test suite, and this manuscript — which is the architectural point of the exemplar. Because every number here is produced by a tested function and regenerated on demand, the prose describes structure and provenance rather than transcribing values that would drift the moment the example methods changed.

5 Conclusion

This paper presented a small, tested domain language for specifying controlled methods, informed by BPL's [Bota Biosciences, 2026] domain-language design for laboratory protocols and generalized to any controlled procedure. It validates a simple proposition: a controlled vocabulary expressed as typed, validated dataclasses — not a parsed text grammar — is sufficient to reproduce BPL's core safety properties at a scope appropriate for a template exemplar.

5.1 Exemplar achievements

Operating as the methods-paper exemplar for the Research Project Template methodology, the project deployed the three foundational pillars:

1. **library**: a controlled vocabulary, a dimensional unit system, four staged validation gates, a deterministic compiler, and four export formats — with no plotting, no file I/O, and (with one declared logging exception) no imports.
2. **integrity**: a zero-mock suite over real fixtures covering the success path and every gate-failure mode, under a >=90% project coverage gate.
3. **knowledge operations**: the correspondence with BPL's pipeline, testing philosophy, and operational rules that keep the library, scripts, and manuscript aligned.

5.2 Technical contributions

5.2.1 Controlled vocabulary in code, not a parser

The hallmark of this exemplar is the design choice it demonstrates: a file's text grammar buys generality this template does not need, while a frozen-dataclass model buys construction-time validation () that a parsed AST would have to re-derive. The controlled vocabulary's discipline lives in the *types*, not in concrete syntax.

5.2.2 Honest handling of trust boundaries

's hash-chain documents its own limit explicitly: it detects in-chain tampering but cannot detect a chain rewritten from record zero. This makes the provenance guarantee a visible, testable property rather than an implied cryptographic guarantee the implementation does not actually provide.

5.3 Key insights

1. **Determinism follows from explicit tie-breaking**: Kahn's algorithm [Kahn, 1962] alone does not guarantee a stable schedule across runs — the ascending- tie-break is what makes reproducible, and sec. 4 checks this live rather than asserting it.
2. **Staged short-circuit avoids misleading errors**: running and against a structurally invalid method would report noise, not signal — short-circuits after the first two gates for exactly this reason.
3. **A controlled vocabulary generalizes by restraint, not by expansion**: introduces no new or : its /// steps and sign-off all draw on the same controlled vocabulary uses — nothing was added to support a second domain.

5.4 Future extensions

This foundation could be extended to:

- **A real -style parser**: add // stages ahead of the existing , reusing every downstream gate and the compiler unchanged.
- **More backends**: a execution target with capability-aware lowering, mirroring BPL's Biomek translation layer.
- **A capability registry**: per-target primitive support declarations, generalizing BPL's registry and report.

5.5 Final assessment

The tree is the canonical reference for how a methods paper — a manuscript whose subject is a methodology — stays synchronized with the code implementing that methodology across rebuilds. The pipeline compiled both worked example methods, wrote , , and , and rendered this markdown together with into PDF.

6 Experimental Setup

This section details the controlled vocabulary, worked examples, and software environment used to produce the results.

6.1 Controlled vocabulary

The DSL's vocabulary is declared once in code and consulted by every gate and the compiler — never re-declared per method:

Module	Declares	Cardinality
	, ,	9 step kinds, 3 targets
	, , the unit table	18 controlled units across 8 dimensions
	The four staged gates	4 gates, fixed order

6.2 Worked examples

() returns 2 methods:

Method	Domain	Target	Notable structure
	Wet-lab bench preparation (BPL's own domain; an original example)		A strict 5-step linear chain with a final step
	Instrument calibration (a non-biology controlled procedure)		Mixed automated / steps and a sign-off step

The second example exists specifically to demonstrate that the controlled vocabulary generalizes beyond wet-lab protocols, as sec. 2 claims.

6.3 Pipeline conditions

The experiment overlay () declares three conditions:

- **declared_method** (reference) — a method constructed directly as Python objects, before any validation gate has run.
- **validated_method** (proposed) — the same method after passing all four staged gates and deterministic compilation to a scheduled .
- **automated_target_variant** (variant) — 's steps recompiled against an execution target, ablating the target-compatibility gate's / step boundary (a method's steps must all be -compatible; an method's steps may be either).

The primary metric is gate pass rate: the fraction of staged-gate evaluations a method's steps satisfy.

6.4 Computational environment

- **Language**: Python 3.12.12 on Darwin arm64 (see root for the supported version range).
- **Core dependencies**: , (declared in); the DSL library itself () has zero third-party dependencies beyond the standard library, with one declared logging exception ().
- **Headless plotting**: the analysis script sets before importing matplotlib.

6.5 Pipeline ordering

The typical analysis order is:

1. — compiles every example method, runs all gates, exports worklist/CSV/Mermaid/JSON per method, demonstrates the provenance hash-chain, and writes , printing each output path for manifest collection.
2. — reads and the analysis outputs, then resolves every generated variable in .
3. PDF rendering reads the resolved manuscript tree so figure paths and prose match the analysis that just completed.

6.6 Relation to results

Result (sec. 4)	Producing function ()	Primary inputs
Compiled-plan summary		
Step-count figure	per method	

Result (sec. 4)	Producing function ()	Primary inputs
Gate pass tally		Each example method
Determinism check	called twice	
Trust-chain verification	/	Demonstration chain in

This table is descriptive documentation only; it is not executed as code during the build.

7 Reproducibility

This section explains how to regenerate every artifact in the study from a clean checkout. The exemplar's reproducibility guarantee is structural: each result is produced by a tested function and a thin script, then injected into the manuscript by generated-variable substitution — never transcribed by hand.

7.1 How to regenerate everything

From the repository root:

Or, end to end via the orchestrated pipeline:

7.2 Generated artifact registry

The analysis script writes the following artifacts under :

Artifact	Produced by
, , ,	+ exporters, for
, , ,	+ exporters, for
	Per-method plan summary, consumed by
	tally across both methods
	/ demonstration chain
	Step-count bar chart
	Every generated-variable value, written by

The tree is disposable and regenerated on every run; it is not the source of truth.

7.3 Determinism

- is deterministic by construction: the plan hash is computed from a canonical, sort-keys JSON payload over , , , and each scheduled step's identifying fields — never from a wall-clock timestamp or a UUID.
- breaks scheduling ties by ascending , so the same object always yields the same step order across processes and platforms.
- Yes — recompiles every example method twice at manuscript-build time and compares hashes live, so this guarantee is checked on every build, not merely asserted once in a test.

7.4 Verification (no hand-transcribed numbers)

Every quantitative claim in sec. 4 is either a generated variable sourced from a live analysis output or registered in for evidence-registry validation. The manuscript intentionally does not hand-transcribe volatile values, so prose and artifacts cannot disagree. Configuration provenance is itself injected: is the SHA-256 of at build time, and records when the variables were generated (honoring for byte-reproducible builds).

8 Scope, Related Work, and Positioning

This section situates the exemplar and states explicit boundaries. The goal is not to compete with BPL's substantially larger full compiler pipeline [Bota Biosciences, 2026] — with its Lark grammar, robot backend, and hash-chained audit/compliance layer — but to show how a minimal, test-backed subset of BPL's domain-language design fits the template's reproducibility and rendering stack [Peng, 2011], generalized from wet-lab protocols to any controlled procedure.

8.1 Domain-specific languages for controlled procedures

Encoding a procedure as a program rather than free text is a long-standing software-engineering pattern: Fowler's treatment of domain-specific languages [Fowler, 2010] frames the general case — a notation restricted to exactly the concepts a domain needs. BPL [Bota Biosciences, 2026] applies this specifically to laboratory protocols, adding biology-native types (reagents, labware, MW-aware concentrations), staged validation, and deterministic compilation to a robot or human execution target. The present manuscript restricts attention to the parts of that design that generalize beyond biology: a controlled vocabulary of step intents, a small dimensional-safety unit system, staged validation gates, and deterministic compilation to a hashed plan.

8.2 What this exemplar does not implement (out of scope)

1. **A text grammar and parser.** BPL parses source through a Lark grammar into a typed AST. This exemplar constructs a directly as frozen Python dataclasses — no concrete syntax, no parser, no AST layer.
2. **Biology-native types.** BPL's unit system includes MW-aware concentration conversions and reagent physical-form metadata (,). This exemplar's implements only the dimensional-safety subset (mass, volume, temperature, time, molar concentration, mass concentration, count, dimensionless) needed to demonstrate the "the type system catches " guarantee — molar and mass concentration are kept as distinct dimensions precisely because no MW-aware conversion exists here to safely mix them.
3. **A robot backend.** BPL lowers intents to Biomek i7 primitives (, ,). This exemplar's has no backend-specific lowering stage; it is a scheduling and gate-compatibility concept only.
4. **Cryptographic audit guarantees.** 's hash-chain is deliberately scoped to the same honest boundary BPL itself claims: a consistency check against accidental corruption, not a tamper-proof guarantee against an actor with write access to the entire chain.
5. **SOP-to-DSL generation.** BPL includes an agentic workflow that translates natural-language SOPs into validated programs. This exemplar's worked examples are hand-authored Python, not generated.

8.3 What this project proves about the template

The validation and compilation steps here are a deliberately small subset of BPL's. The **non-standard** contribution is procedural: the same tested functions in back the analysis script, the test suite, and this manuscript, so the compiled-plan table and the figure always refer to the same code. That pattern — and the specific generalization from a biology-only domain language to a domain-neutral one — is what downstream projects should copy, whether the controlled procedure is a wet-lab protocol, an instrument calibration sweep, or a computational pipeline.

8.4 Explicit limitations

1. **Two worked examples**: and exercise every gate-success path but are not a corpus; gate-failure coverage instead lives in 's dedicated fixtures.
2. **No backend lowering**: selects a compatibility class, not a concrete execution backend; no robot or simulation runtime exists in this exemplar.
3. **Unit table, not a full unit library**: eight dimensions and a fixed unit table, not a general-purpose dimensional-analysis library like .
4. **No persistence layer**: 's hash-chain lives in memory for the duration of one script run; this exemplar does not implement durable storage for a real audit trail.

These limitations are intentional: they narrow the surface so that the reproducibility concerns — tested functions, a thin script, and generated-variable-injected prose — remain visible rather than buried under a compiler implementation at BPL's full scale.

9 References

Bibliography lives in and is read by Pandoc during PDF render. The build pipeline invokes Pandoc with , so every citation in the manuscript is rewritten to the appropriate // LaTeX command and resolved against the bib file.

To validate that is syntactically clean and contains the required fields per entry type:

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